

Mateusz Płoszaj-Mazurek

Building researcher · Daylight, building regulation and computation

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(<https://github.com/Curiosit>)

Researcher at the Faculty of Architecture, Warsaw University of Technology. My PhD and the 2020 and 2024 articles in *Energies* developed machine-learning, parametric and BIM-based methods for low-carbon architectural design. My current work moves the same approach to daylight: fast surrogate models that predict climate-based daylight metrics from geometry, and parametric studies of how daylight requirements perform.

Publications

A workflow for LLM-assisted IFC data enrichment to support circular reuse of large-panel building systems in Poland

with A. M. Tofiluk · *Journal of Building Engineering* 120, 115397 (2026) · doi · open access

Green Data Centres: Sustainable Solutions with Green Energy and Green-Blue Infrastructure

M. Grochulska-Salak, E. Maciejewska, P. A. Bujak et al. · *Energies* 18(24), 6592 (2025) · doi

Artificial Intelligence and Digital Tools for Assisting Low-Carbon Architectural Design: Merging the Use of Machine Learning, Large Language Models, and Building Information Modeling for Life Cycle Assessment Tool Development

with E. Ryńska · *Energies* 17(12), 2997 (2024) · doi

Methods to Optimize Carbon Footprint of Buildings in Regenerative Architectural Design with the Use of Machine Learning, Convolutional Neural Network, and Parametric Design

with E. Ryńska and M. Grochulska-Salak · *Energies* 13(20), 5289 (2020) · doi **136 citations**

Machine Learning-Aided Architectural Design for Carbon Footprint Reduction

Builder 276 (7/2020), *Builder Science*, pp. 35–39 · doi

Digital methods of supporting architectural design process and the analysis of the carbon footprint of buildings

PhD thesis · Warsaw University of Technology (2022)

W kierunku architektury prośrodowiskowej – sztuczna inteligencja jako wsparcie architektonicznego procesu projektowego

Chapter in: *Architektura i urbanistyka wobec sytuacji kryzysowych*, eds K. Solarek, C. Głuszek · Oficyna Wydawnicza Politechniki Warszawskiej (2022), pp. 63–75

Whole Life Carbon Roadmap for Poland: How to decarbonise the built environment by 2050

A. Kuczera, M. Płoszaj-Mazurek, A. Hadziivanov et al. · *PLGBC with WorldGBC and EBRD* (2021), 52 pp. · doi

Budownictwo drewniane a architektura zrównoważona

Chapter in: *VII Forum Holz bau Polska HBP2023*, ed. M. Wnorowska · *Forester Communication* (2023), pp. 215–222

Parametric Optimisation of Carbon Footprints

Chapter in: *Regenerative Design in Digital Practice. A Handbook for the Built Environment*, eds E. Naboni, L. Havinga · *COST Action RESTORE* (2019), pp. 258–263

Parametryczna optymalizacja śladu węglowego budynków

Chapter in: *Miasto dla ludzi – miasto jutra*, ed. B. Gronostajska · *Politechnika Wrocławska* (2019), pp. 303–315

Prescriptive and Performance Based Daylighting Regulation as an Architectural Experiment: Parametric Simulation of Polish Residential Standards

Architecture of Challenges: Experiment in Space – the Space of Experiment, ed. A. M. Wierzbicka · Warsaw University of Technology (2026), pp. 160–161

Implementing Building Energy Modelling in a Master's Level Architectural Design Course: Preliminary Insights from a Pilot Study

with M. Gołębiowski, M. Pierzchański · *Architecture of Challenges* (2026), pp. 62–63

Predicting Daylight in Milliseconds: A Computational Machine Learning Surrogate Model for Early-Stage Urban Housing Design Simulation. Artificial Dataset Generation

Poster · Young Scientist Forum "200 Młodych na 200-lecie Politechniki Warszawskiej" (2026)

Artificial intelligence for reducing carbon footprint of buildings. A comprehensive approach using convolutional neural networks and large language models in architectural design optimization
 Redefining Cities in View of Climatic Changes 2023: Abstracts · Warsaw University of Technology (2023), pp. 96–97

Artificial intelligence and machine learning for low carbon architecture. Prototype carbon estimation tool development case study
 Structures and Architecture. A Viable Urban Perspective?, eds P. J. S. Cruz, M. Frier Hvejsel · CRC Press (2022), vol. 2, pp. 345–346 · doi

Parametric Optimization of Carbon Footprint
 YRSB19: iiSBE Forum of Young Researchers in Sustainable Building, Prague · Czech Technical University (2019), pp. 165–174

Applying Machine Learning in Integrated Architectural Design for Life Cycle Carbon Footprint Reduction
 Redefining Cities in View of Climatic Changes 2019 · Warsaw University of Technology (2019), pp. 117–118

Analytic radiometric descriptors enable geometry-flexible per-sensor daylight prediction without learned spatial encodings [preprint](#)
 SSRN (2026) · code github.com/Curiosit/OmegaFlux · software doi:10.5281/zenodo.21781259

Akademik na Solcu, Warsaw
 Project publication, with M. Czechowicz · Architektura-Murator 4/2025, pp. 54–64

Przedszkole pasywne, Reguły
 Passive House kindergarten, project publication, with D. Sibińska, J. Sokołowski et al. · Architektura-Murator 5/2023, pp. 20–34

Prefabrication and sustainable design: a three-part series
 with A. M. Tofiluk · Builder 4/2022, 7/2022 and 10/2022 · plus "Prefabrykacja w architekturze zrównoważonej", Zawód: Architekt 9–10/2022

Nowe wyzwanie dla polskiej architektury
 J. Biernacka, P. Biernacki, P. Dowgiało et al. · Architektura-Murator 3/2019, pp. 30–32

Selected projects

2025–26	OmegaFlux Per-point daylight prediction from geometric view descriptors with gradient-boosted trees, exported to ONNX and run in the browser. Dataset generation, held-out validation against Radiance, transfer to unseen room shapes. curiosit.github.io/OmegaFlux (https://curiosit.github.io/OmegaFlux/)
2021	Whole Life Carbon Roadmap for Poland PLGBC, WorldGBC, EBRD Co-author, with Alicja Kuczera, of the national roadmap for decarbonising the built environment by 2050.
2018–now	Multi-criteria building optimization Evolutionary and multi-objective optimization of energy, daylight and carbon in parametric models; taught as a course module.

Grants and awards

2024–25	Principal investigator, IDUB research grant · Circular economy in architecture: reuse of prefabricated elements and digital design techniques · Warsaw University of Technology, CPR-IDUB/238/Z01/2024
2024	First prize, Best Mashup Project · AEC Hackathon, Copenhagen
2024	Rector's team award for organisational achievements (first degree) · Warsaw University of Technology, academic year 2022/23
2023	Rector's team award for teaching achievements (third degree) · Warsaw University of Technology, academic year 2021/22
2023	Competition win: energy-efficient multi-storey housing in Poland · Winning entry by Studio Widoki, with Bjerg Arkitektur as consultant · Carbon footprint calculations by Bjerg Arkitektur: prefabricated timber, about 6 kg CO ₂ e/m ² per year over 50 years
2022	PLGBC Green Building Awards: best green building without certification · Passive House kindergarten in Podkowa Leśna, Bjerg Arkitektur; award received on behalf of the practice. Also a distinction for an educational building.
2022	Distinction for the doctoral thesis · Innowator Mazowsza competition

Talks and panels

2023	Pro-environmental design: training sessions for architects · Saint-Gobain · two sessions, Jura Krakowsko-Częstochowska and Żnin, October 2023
2023	ESG Digital Leader Academy · Proptech Foundation · development programme for managers implementing ESG strategy with digital tools
2023	Lublin on its way to Twin Transition: Danish–Polish workshop · Lublin, 13 April 2023
2026	Discussion panels · IV International Conference Architecture of Challenges: Experiment in Space, Warsaw, 29–30 June 2026
2025	Session "Architecture of Community" and discussion panels · III International Conference Architecture of Challenges: New European Bauhaus– Building Community, Warsaw, 30 June – 1 July 2025
2025	Decarbonising buildings in the climate crisis: policy, practice and the investor's view · PLGBC Green Building Summit
2024	Digital and business challenges for developers in 2024: non-financial reporting and carbon footprint · Building & Human Summit, Łódź, March 2024
2023	How to choose software for managing a building's carbon footprint · Webinar, Proptech Foundation, 13 December 2023
2023	Energy resilience as a priority in urban and architectural design · III Kongres Regeneracja Miast, Łódź, 31 May 2023
2023	"Trochę architektura, trochę biologia" · Młodzi do Łodzi, conference of Architektura-Murator, Łódź, 26–27 May 2023
2021	Conversation with a Sweco sustainability and BIM lead · Miasta Przyszłości, 4th edition, online
2021	Launch of the Whole Life Carbon Roadmap for Poland · Webinar with PLGBC, EBRD and WorldGBC, 9 June 2021
2018	"Zero Carbon. Road to sustainable home" · 8th Green Building Symposium, PLGBC
2025	Sztuka vs. Sztuczna Inteligencja. Algorytm z duszą i emocjami – to tylko kwestia czasu? · 22nd Warsaw Art Fair (Warszawskie Targi Sztuki)
2025	Early-phase building carbon footprint optimization with Carbon Designer 3D · Carbon Experts Summit 2025
2023	Special presentation, "The role of steel in the future" · V Dzień Stali, BUDUJ ZE STALI, Warsaw, 28 November 2023
2023	Timber building and sustainable architecture · VII Forum Holzbau Polska, Jachranka, 25–26 April 2023
2023	Low-carbon sustainable architecture and the use of BIM · Mistrzowie Krajobrazu 2023, online, 21 April 2023
2022	Bjerg Arkitektur's approach to climate in construction · Kongres Regeneracja Miast, Łódź, June 2022
2021	How prefabrication lowers the carbon footprint of buildings · Sztuka Architektury, online, 27 May 2021

Experience

current	Researcher and lecturer Faculty of Architecture, Warsaw University of Technology - Daylight regulation, simulation and surrogate-model research; see publications. - Teaches Grasshopper and multi-criteria building optimization; coordinates a lecture cycle on public buildings; runs a second-year design studio. - Designs parametric experiments and evaluation frameworks: thousands of simulated room variants, held-out test splits, convergence checks.
2024–2026	Software engineer, interim product owner One Click LCA, Helsinki · remote Built features for Carbon Designer 3D, an early-design tool for architects (TypeScript, React, Three.js, AG Grid).

	2014–2024	Architect, Certified Passive House Designer Bjerg Arkitektur · Hjørring 2014–16, Warsaw 2016–24 - Single- and multi-family housing, schools, kindergartens and offices, from 100 m² to 15,000 m². - All design phases: concept, permit and construction documentation, and architectural supervision on site. 2022–2024: architecture, sustainability and software consultant, building in-house design tools; co-author of the Polish whole-life carbon roadmap.
Teaching	7 years	Introduction to Grasshopper Parametric modelling and multi-criteria optimization of energy, daylight and carbon.
	ongoing	Computing Architecture for Sustainable Environment Seminar for fourth- and fifth-year students, including daylight simulation.
	ongoing	Design studio and lecture cycle Second-year studio on small public architecture; coordinator of the semester-seven lectures on public buildings; English-language Solar Seminar.
	2021–	Techniques Integration, ASK Co-taught course integrating technical and environmental design.
Service	2026–	Guest co-editor, <i>Buildings</i> (MDPI) · Special issue on performance-responsive architecture and urban design
	2025–	Reviewer · Energy and Buildings, Automation in Construction, Buildings, Energies, Architecture, Digital
	2024–26	Jury member · National competition for the best pro-environmental diploma project; judge-rapporteur in 2025
	2023–26	Conference committees · Organising committee, Architecture of Challenges IV (2026); scientific and organising committees, Redefining Cities in View of Climatic Changes (2023)
	2018–23	Editor, catalogue of the best diploma projects · Faculty of Architecture, Warsaw University of Technology, five volumes, two co-edited with A. Grabowska
	current	Auxiliary supervisor · Doctoral candidate at Warsaw University of Technology
	current	Member, Sustainable Architecture Circle, SARP
Education	2022	PhD in Architecture Warsaw University of Technology Digital methods of supporting the architectural design process and the analysis of the carbon footprint of buildings: parametric optimization and machine-learning methods for design.
	2014–2016	MSc in Architecture Aalborg University, Denmark
	2023–2024	Certifications Master BIM Software Developer (That Open Company, 2024) · Harvard CS50x (2023) · Certified Passive House Designer
Skills	Building domain	Climate-based daylight (sDA, ASE, UDI, EN 17037), insolation, Polish and Danish building regulations, Passive House, building carbon footprint
	Methods	Evolutionary and multi-objective optimization, surrogate modelling, gradient-boosted trees, dataset and validation design, parametric experiments
	Design tools	Rhino and Grasshopper, Ladybug and Honeybee, Radiance, THERM, IFC and BIM, That Open Company tools
	Code	Python (NumPy, pandas, scikit-learn, XGBoost, ONNX), TypeScript, React, Three.js, Git, CI/CD
	Languages	Polish (native), English (fluent)